Investigating Ideological Bias in Large Language Models:

A Comparative Study of Role and Language Effects

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Abstract

Contents

Chapter 1

Introduction

1.1 Motivation

Large Language Models (LLMs) such as GPT [1] and LLaMA [2] are now deployed worldwide and used in many languages. As these systems mediate information access and public discourse, a central concern is whether they present the same opinions and reasoning to users across countries and languages—or whether they exhibit ideological or cultural biases [3] [4]. Understanding these effects is essential for trustworthy, fair, and transparent AI [5].

1.2 Problem Statement

Existing audits have shown that large language models (LLMs) can encode biases along several axes, including social, linguistic, and representational dimensions [6] [7] [8]. However, most empirical studies remain limited to a single country, a single demographic axis, or rely heavily on manual prompt probes and human judgments [9]. Less attention has been paid to *cross-country, cross-language ideological behavior* in contemporary chat models, and to disentangling the influence of *role framing* (e.g., "answer as a citizen of X") from *language framing* (answering in X's native language). Recent work has begun to examine political and national biases in LLMs [10] [11] [12], but reproducibility remains a challenge: manual inspection is often subjective and difficult to replicate. This motivates the need for a multilingual, reproducible evaluation that targets *stance and reasoning* on policy-relevant topics, while reducing dependence on ad-hoc human judging.

1.3 Research Objectives and Questions

Objectives. (i) Build a reproducible framework to evaluate ideological bias across roles and languages [10]; (ii) analyze both stance alignment and the semantic similarity of justifications [8]; (iii) compare patterns across model families (GPT vs. LLaMA).

Research Questions.

1. Do LLMs exhibit ideological bias across countries and policy topics?
2. Are model outputs more sensitive to *role* framing or to *language* framing?
3. How similar are role-based and native-language responses for the same country?
4. Are the observed effects consistent across model families (GPT vs. LLaMA)?

1.4 Methodology (overview)

We adopt a multilingual, stance-based audit designed to minimize subjectivity and maximize reproducibility.

- **Data.** We use a stance-labeled argument corpus (`arguments-training.tsv`, SemEval ValueEval / Webis-ArgValues) containing thousands of premises, conclusions, and gold stances ("in favor"/"against") [13][14].
- **Topic focus.** To concentrate on policy-relevant debates, we apply BERTopic [15] to the corpus and retain eight ideologically salient themes (e.g., nuclear weapons, economic sanctions, religion/atheism, compulsory voting, libertarianism, etc.).
- **Prompting conditions.** For each argument we elicit (a) **role-based** completions by setting the system persona to "answer as a citizen of {America, Iran, Israel, Ukraine, Germany, Italy, France, China}" plus a neutral assistant; and (b)

native-language completions by translating the same instruction into each country's primary language (no persona). Our setup extends prior work that has studied role/persona framing or cross-lingual bias in isolation [10][11][9].

- **Models and outputs.** We run GPT and LLaMA. Each response is constrained to a JSON object with a binary stance (`Agree/Disagree`) and a short justification; refusals and formatting errors are logged, in line with other prompt-based evaluation frameworks [8].

- **Metrics.** (1) *Agreement ratio*: compares model stance to gold labels per row and aggregates by topic/condition (with a neutral-aware denominator for LLaMA when needed), inspired by stance-alignment metrics in prior bias audits [16][7]. (2) *Embedding similarity*: uses LaBSE [17] to embed justifications and computes cosine similarity on matched arguments for three comparisons: Role↔Assistant, Native↔Assistant, and Role↔Native.

This design targets ideological stance and reasoning, operates natively across languages, and avoids ad-hoc manual judging.

1.5 Thesis Structure

The remainder of the thesis is organized as follows. *Background* introduces bias taxonomies and evaluation paradigms relevant to our study. *Related Works* surveys empirical audits of LLM bias (stereotypes, religion, gender), multilingual behavior, and role/persona effects, and contrasts prompt-probe methods with stance-based evaluation. *Methodology* details the dataset, topic modeling, prompting setup, models, and metrics. *Analysis of Results* reports agreement and similarity outcomes across topics, countries, and models. *Limitations and Conclusion* discusses constraints of our approach and outlines directions for future work.

Chapter 2

Background

2.1 Large Language Models (LLMs)

Over the past decade, natural language processing (NLP) has undergone a major paradigm shift, moving from static word embeddings to large-scale pretrained language models.

Static word embeddings. Early approaches such as Word2Vec and GloVe mapped each word to a single fixed vector in a continuous space [18][19]. These methods captured semantic similarity (e.g., "king – man + woman ≈ queen") but could not represent polysemy, since each word had only one embedding regardless of context.

Contextual embeddings. The next step was contextualized representations, where the embedding of a word depends on its surrounding text. Models such as ELMo and BERT introduced this capability [20][21]. This shift enabled richer representations and significantly improved performance across many NLP tasks.

Autoregressive language models. In parallel, autoregressive models such as GPT adopted a left-to-right prediction strategy, generating text one token at a time conditioned on the previous context [22][23]. Scaling these models to billions of parameters demonstrated strong zero- and few-shot learning abilities. More recently, the LLaMA family introduced by Meta has followed the same decoder-only autoregressive design, offering competitive performance at smaller parameter scales by leveraging efficient training and curated public datasets [2][24].

Transformer architecture. Transformer-based models can be broadly divided into encoder-style (e.g., BERT [21]) and decoder-style (e.g., GPT [22][23], LLaMA [2]). The transformer itself [25] relies on self-attention to capture long-range dependencies, avoiding the sequential bottlenecks of recurrent and convolutional networks (RNNs, CNNs). Decoder-only transformers have become the foundation for generative LLMs, while encoder-only models remain widely used for embeddings and retrieval tasks.

Instruction tuning. More recently, models have been fine-tuned with human feedback to better follow natural instructions. InstructGPT and ChatGPT exemplify this trend, aligning LLM behavior with user intent through reinforcement learning from human feedback (RLHF) [26]. Instruction-tuned variants of LLaMA (e.g., LLaMA-2-Chat) extend this paradigm in the open-source domain, making them valuable for research into model alignment and bias.

LLMs are not only technical achievements but also powerful social actors. Their outputs influence online information flows, education, journalism, and decision-making at scale [27][5]. As a result, questions about fairness, bias, and accountability have become central: models can reproduce and even amplify societal stereotypes, political leanings, and ideological framings. Understanding these behaviors is thus essential to ensure responsible deployment of LLMs in sensitive domains.

2.2 Bias in NLP and LLMs

The concept of bias in natural language processing (NLP) and large language models (LLMs) has been widely discussed across computer science, linguistics, and ethics. In this thesis, we adopt a broad definition of bias as any systematic deviation in model behavior that disproportionately favors, disadvantages, or misrepresents particular social groups, viewpoints, or languages [5][27].

Types of bias. Several dimensions of bias are relevant for NLP systems:

- **Social and representational bias.** These biases occur when a model reflects or amplifies stereotypes associated with gender, race, age, religion, or profession.

For example, occupation-related prompts may reproduce gender stereotypes, or religious terms may be disproportionately associated with negative sentiment [9][7][16].

- **Ideological bias.** Models may encode political leanings or preferences for particular policy positions, subtly framing issues such as international conflict, economic sanctions, or social freedoms in ways that reflect the biases of their training data [10][28][12]. This type of bias is especially important when LLMs are used in settings where neutrality is expected, such as journalism, education, or policy debate.

- **Linguistic and cross-lingual bias.** Since many LLMs are primarily trained on English or high-resource languages, their behavior can vary significantly across languages. Cross-lingual evaluations show that low-resource languages often yield less accurate, less safe, or differently framed responses [11][10]. In addition, the language of a prompt itself can introduce ideological skew, independent of content, due to cultural or linguistic associations.

Implications. These forms of bias highlight that LLMs are not neutral information processors but active participants in shaping discourse. Standard surveys emphasize that mitigating bias is not only a technical challenge but also a social one, requiring awareness of how models interact with human values and institutions [5][27]. In this thesis, we focus specifically on *ideological bias across roles and languages*, which has received comparatively less attention than stereotype or demographic biases.

2.3 Stance Detection and Argument Mining

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2.4 Bias Measurement in NLP

A wide range of methods have been proposed to quantify bias in language models. Early approaches drew inspiration from psychology, in particular the *Implicit Association Test* (IAT), which measures how quickly people associate target concepts with attributes [29].

In NLP, this idea was adapted into the *Word Embedding Association Test* (WEAT), which evaluates whether vector representations of words (e.g., "man" vs. "woman") are more closely associated with stereotypical attributes (e.g., "career" vs. "family") [6].

Subsequent extensions generalized these ideas to contextualized embeddings and sentence encoders. The *Sentence Encoder Association Test* (SEAT) evaluated biases in sentence-level models [30], while large-scale benchmarks such as *StereoSet* [7] and *CrowS-Pairs* [16] introduced datasets for testing gender, profession, race, and religion biases in pretrained LLMs.

More recent work has explored fairness metrics that go beyond associations, such as pairwise and multi-group comparison metrics, which compare outputs across demographic groups at a system level [31]. These approaches aim to capture not only representational bias but also extrinsic disparities in model behavior.

Finally, prompt-based evaluation methods have become prominent in the LLM era. Instead of probing embeddings directly, these methods construct diagnostic prompts and analyze the generated completions. For example, Aowal et al. [8] introduce both human- and model-level self-diagnosis techniques for prompt evaluation, while Haller, Aynetdinov, and Akbik [28] propose a fine-tuned model explicitly designed to surface political, geographic, and demographic biases.

Together, these strands of research show the progression from embedding-level association tests to prompt-based behavioral audits. Our approach follows this trajectory by introducing stance- and reasoning-focused measures (agreement ratios and embedding similarities) that are reproducible and directly applicable to multilingual, generative models.

2.5 Embeddings and Semantic Similarity

A central concept in modern NLP is the use of embeddings: vector representations of words, sentences, or documents in a continuous space.

Word embeddings. Early models such as Word2Vec [18] and GloVe [19] mapped each word to a single static vector, capturing distributional similarity but failing to account for context. For example, the word "bank" would have the same representation whether referring to a riverbank or a financial institution.

Contextual embeddings. To address this limitation, contextualized representations were introduced. Models like ELMo [20] and BERT [21] produce embeddings that depend on the surrounding text, enabling richer and more accurate semantic representations. These advances marked a major shift in NLP, underpinning many downstream improvements.

Sentence embeddings. Building on contextual models, sentence-level encoders were developed to represent entire utterances or documents as dense vectors. Such models enable semantic similarity comparisons through measures such as cosine distance. They are widely used in clustering, retrieval, and argument similarity tasks [32].

Cross-lingual embeddings. For multilingual applications, cross-lingual models align semantically similar sentences across languages into a shared embedding space. Multilingual BERT (mBERT) extended BERT to 104 languages, though with uneven performance across resource levels [21]. More recent models like LaBSE (Language-Agnostic BERT Sentence Embedding) provide stronger cross-lingual alignment, supporting over 100 languages with consistent performance [17].

Relevance to this thesis. Embedding similarity provides a scalable way to compare not only the stances of LLM outputs but also the *reasons* behind them. By embedding justifications and computing cosine similarity across role-based, native-language, and assistant completions, we can quantify ideological consistency at the level of argumentation rather than surface stance alone. This enables reproducible, multilingual comparisons of reasoning that go beyond manual interpretation.

Chapter 3

Related Works

3.1 Audits of Bias in LLMs

Much of the literature has examined **social and representational biases**, particularly around gender, race, and religion. For example, Gross [9] demonstrate how ChatGPT responses reproduce gender stereotypes, while Abid, Farooqi, and Zou [33] identify systematic anti-Muslim bias across GPT-3 outputs.

Beyond demographic bias, some works investigate **political and ideological bias**. Urman and Makhortykh [10] conduct a cross-lingual analysis of political leanings and misinformation in ChatGPT, Bard, and Bing Chat, while Haller, Aynetdinov, and Akbik [28] introduce a fine-tuned model explicitly designed to surface ideological positions. Similarly, Rogers and Zhang [12] analyze how LLMs portray the Russia–Ukraine war in Chinese social media, finding a systematic bias toward neutrality.

A smaller body of research focuses on **censorship and neutrality**. For instance, Yang and Roberts [34] examine how restrictions in online encyclopedias shape NLP training data, and Du, Wei, and Ji [35] study the downstream implications of censorship for LLM behavior. These findings highlight that bias can also emerge from absences and silences in the training corpus.

3.2 Cross-Lingual and Cross-Cultural Bias

Bias does not manifest uniformly across languages. Early studies documented **nationality bias** in text generation [11], while more recent audits show that models like ChatGPT behave differently depending on the input language [10]. However, most of these works compare languages without disentangling the effect of *persona framing* (e.g., "answer as a German citizen") from *native-language framing* (answering in German). To our knowledge, few studies explicitly test this distinction, which constitutes a core gap addressed in our thesis.

3.3 Bias Detection Datasets

A variety of benchmark datasets have been introduced to study bias in NLP systems. CrowS-Pairs [16] and StereoSet [7] are widely used to probe stereotypical associations in masked LMs through controlled sentence pairs or context association tests. Other resources such as WinoBias [36] and GAP [37] focus specifically on gender bias in coreference resolution. By contrast, BOLD [38] extends bias analysis to *open-ended generation tasks*, prompting models to produce unconstrained outputs such as stories or descriptions, which can reveal subtler discursive biases beyond word-level associations.

Beyond stereotypes, argument- and stance-oriented datasets provide more structured evaluations. The Webis-ArgValues corpus from the SemEval ValueEval shared task [13] contains thousands of arguments with gold stances ("in favor" vs. "against") across diverse policy issues, offering a controlled way to study opinionated text. Similarly, OpinionGPT [28] leverages Reddit-based corpora to surface explicit political and demographic biases.

Our work is methodologically closest to BOLD in analyzing *open-ended generations*, but we introduce additional structure by constraining outputs into stance labels and short justifications. This hybrid format combines the richness of open-ended text with the reproducibility of stance-based annotation, enabling systematic cross-country and cross-language comparisons that go beyond existing stereotype-focused resources.

3.4 Evaluation Metrics and Frameworks

Bias measurement methods range from embedding-level association tests to prompt-based audits. Early work adapted psychological tests into the Word Embedding Association Test (WEAT) and its sentence-level counterpart, SEAT [6, 30]. Extrinsic fairness metrics later extended these approaches, introducing pairwise, background, and multi-group comparisons to evaluate disparities in system outputs [31].

In the LLM era, prompt-based approaches have gained prominence. Aowal et al. [8] propose human- and model-level self-diagnosis methods, while Haller, Aynetdinov, and Akbik [28] show how instruction-tuned models can reveal political and demographic biases. However, many such approaches rely heavily on manual inspection, which limits reproducibility.

3.5 Gap Analysis

From this review, three gaps become apparent:

1. Most studies focus on a single demographic axis, a single country, or a single language.

2. Cross-country and cross-language analyses remain rare, and the specific distinction between *role framing* and *language framing* has been largely overlooked.

3. Existing methods often emphasize stereotypes or associations, rather than systematically comparing stance and reasoning across conditions.

Our contribution addresses these gaps by (i) introducing a reproducible stance-based framework, (ii) evaluating both agreement ratios and semantic similarity of justifications, and (iii) comparing patterns across role-based and language-based prompts for multiple countries and two model families (GPT and LLaMA).

3.6 Visual Summary

Table 3.1 provides a high-level summary of related studies, contrasting their bias type, language coverage, methods, and limitations.

Study	**Bias Type**	**Language(s)**	**Method**	**Limitations**
StereoSet [7]	Stereotype (gender, race, profession, religion)	English	Association tests with contextual LMs	Focused on stereotypes; single language
CrowS-Pairs [16]	Stereotype (US-centric)	English	Paired-sentence probes	Narrow cultural scope
Gender Bias in ChatGPT [9]	Gender stereotypes	English	Prompt-based audit	Manual inspection; limited scope
OpinionGPT [28]	Political, demographic	English (Reddit)	Instruction-tuned model	Domain-specific; limited generality
Silence of the LLMs [10]	Political, misinformation	Multilingual	Prompt-based, cross-lingual	Language effects not disentangled from role framing
Russia–Ukraine War [12]	Political neutrality	Chinese social media	Content analysis of LLM outputs	Single case; single conflict

Table 3.1: Summary of representative related works, showing their focus, scope, and limitations relative to this thesis.

Chapter 4

Methodology

4.1 Dataset and Topic Modeling

4.1.1 Dataset

We use `arguments-training.tsv` from SemEval 2023 Task 4: ValueEval [13], based on the Webis-ArgValues-22 corpus [14], containing **5,393** stance-labeled arguments with fields `ID`, `premise`, `conclusion`, and `stance` (in favor / against). Stance labels allow auditable, per-row comparison between model outputs and human annotations.

4.1.2 Why this corpus

Our evaluation requires (i) many arguments per theme, (ii) explicit stance labels, and (iii) coverage of sensitive socio-political topics. ValueEval satisfies these simultaneously, enabling reproducible bias measures rather than ad-hoc human judgments.

4.1.3 Topic modeling with BERTopic

Because the raw topics are heterogeneous and often fine-grained, we cluster arguments with **BERTopic** [15], using transformer embeddings (OpenAI `gpt-3.5-turbo` [1]), UMAP for dimensionality reduction, and HDBSCAN for clustering (default parameters unless noted). BERTopic produced **88** topics; we manually inspected keywords and exemplars and retained **8** ideologically salient themes:

ID	Theme
3	Nuclear weapons & war
4	Sex selection & gender
15	Economic sanctions & countries
17	Legalizing prostitution
18	Multi-party system
22	Atheism & religion
36	Compulsory voting
40	Libertarianism & government

We use BERTopic's `topic_info.csv` (topic size, top words, representatives) to audit clusters; sanity checks include the intertopic distance map (reported as a figure in the Appendix).

4.2 Prompting and Response Collection

4.2.1 Prompt design

For each argument we request a JSON object with a binary stance and a short justification:

```
{"idea": "Agree", "why": "this is my reason"}
```

We run two settings:

1. **Role-based:** system role = {*America, Iran, Israel, Ukraine, Germany, Italy, France, China*} and a neutral *assistant*.
2. **Native-language:** same instruction translated to the country's primary language; no role persona.

To reduce parsing errors we enable JSON-enforced replies where supported `response_format={"type":"json_object"}`.

4.2.2 Models

We query two families:

- **GPT-4o-mini** (OpenAI; JSON enforcement available [39]),
- **LLaMA 3.1-70B**, a successor in the LLaMA family [2]

4.2.3 Translation and label normalization

Native-language prompts are translated using the `deep-translator` Python library [40], which provides programmatic access to Google Translate (*DeepL* was not used since it lacks support for Persian and Hebrew). Returned stance tokens are normalized to canonical `Agree`/`Disagree` labels per language. The explicit mapping table is included in the Appendix for auditability.

4.2.4 Logging and error handling

For each call we store: model, condition (role/native), topic ID, argument ID, raw text, parsed JSON, and flags for API/JSON errors, soft refusals, or empty output. Error rows are excluded from metric denominators but reported separately.

4.3 Bias Measures

4.3.1 Agreement ratio (stance-level)

For topic t and condition r:

$$\text{AgreeRatio}(t, r) = \frac{\#\text{agree}}{\#\text{agree} + \#\text{disagree}}$$

Agreement means the model label matches the gold stance for that row (e.g., gold=*against*, model=`Disagree`).

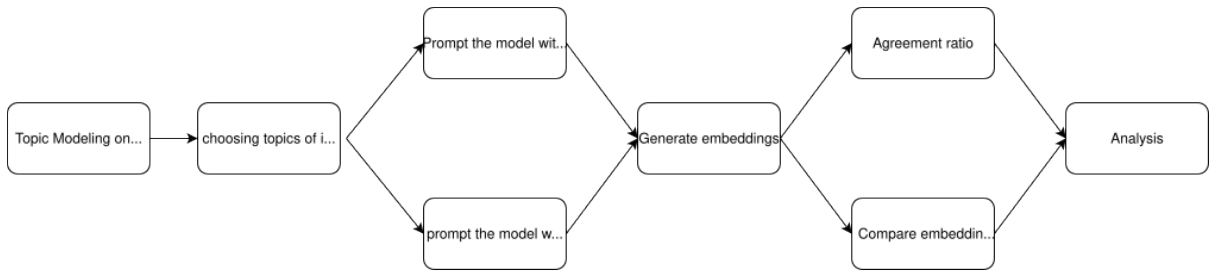

Figure 4.1: Methodology pipeline: BERTopic selection, prompting (role/native), stance agreement, and embedding similarity.

LLaMA adjustment. When `Neutral`/`Neither`/`Partial` occurs:

$$\text{AgreeRatio}^{\text{LLAMA}}(t, r) = \frac{\#\text{agree}}{\#\text{agree} + \#\text{disagree} + \#\text{neutral}}$$

where `Partially Agree` counts as `agree`; `Neither/Neutral` counts as `neutral`.

4.3.2 Embedding similarity of justifications

To compare the *reasons* (beyond labels), we embed `LLM_reason` with multilingual **LaBSE** [17] (768-D) and compute cosine similarity on matched rows between:

1. Role-based vs. Assistant,
2. Native-language vs. Assistant,
3. Role-based vs. Native-language.

Per topic/comparison we report mean, standard deviation, and n.

4.4 Analysis Plan (per model)

We run the three pairwise comparisons above, and additionally (i) summarize JSON error/refusal rates by condition, and (ii) examine topic sensitivity (which themes show the largest language-vs-role gaps). All figures/tables are generated from logged artifacts for full reproducibility.

Chapter 5

Analysis of Results

5.1 Dataset Exploration

Examples. Table 5.1 shows representative rows from the corpus. Note that the stance label is annotated *relative to the conclusion* ("in favor of" vs. "against"). Some conclusions (e.g.,on nuclear energy) appear multiple times with diverse conclusions, and not all conclusions fall within the eight ideological themes we analyze later. This motivates our use of topic modeling on the *premise* text to obtain higher-level themes.

Distinct conclusions. The dataset contains **332** distinct conclusions. The number of arguments per conclusion is highly variable, ranging from as few as **1** to as many as **114**. This indicates that the dataset is not well balanced across different conclusions.

Stance distribution. Figure 5.1 reports the overall distribution of stance labels relative to the conclusion. The corpus is moderately balanced, with "in favor of" slightly more common.

5.2 Topic Modeling Results

After applying topic modeling with **BERTopic**, the output is stored in a CSV file where each row corresponds to a discovered cluster. Each row contains the following fields:

Argument ID	Conclusion	Stance	Premise
A01009	We should fight for the abolition of nuclear weapons	against	nuclear weapons help keep the peace in uncertain times.
A05040	We should legalize sex selection	against	we should not do this because other countries like China already did and it caused an epidemic where there is a huge imbalance of males to females which disrupts the population.
A05057	We should subsidize Wikipedia	against	subsidizing Wikipedia could be a back gateway to some form of governmental control, debasing the validity of the articles and information provided.
A05048	We should ban fast food	against	fast food is sometimes the only way poor people can feed their families cheaply. If you ban it then you are taking away someone's only meal.
A05089	We should ban missionary work	against	missionary work is beneficial for those receiving aid.
E04002	We need to invest in nuclear energy	against	Nuclear energy is too expensive and hence economically not viable. According to researchers at RethinkX, a clean energy disruption in the next ten years may leave investments in oil, coal, and nuclear stranded.
E01030	Gas and nuclear power should be supported with money from European countries	against	Continuing to invest in environmentally damaging and immensely expensive nuclear power plants is a waste of resources that could be used for sustainable transformations.
A05017	cosmetic surgery is unnecessary and dangerous.	in favor of	We should ban cosmetic surgery

Table 5.1: Sample arguments with conclusions, stance labels and premises.

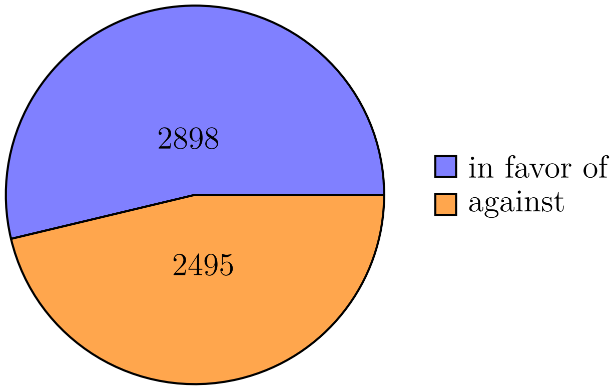

Figure 5.1: Distribution of stance labels relative to conclusions.

`Topic` (cluster ID), `Count` (number of arguments assigned to that cluster), `Name` (an auto-generated label from the most salient keywords), `Representation` (the highest c-TF–IDF keywords), and `Representative_Docs` (sample premises from the cluster).

Table 5.2 reports several clusters that were selected as the topics of interest for this project, including nuclear weapons, sex selection, prostitution, economic sanctions, multi-party systems, atheism, compulsory voting, and libertarianism.

Figure 5.2 illustrates the top seven most frequent clusters together with their representative keywords and their relative importance scores (as assigned by BERTopic). This provides intuition about the main lexical features that characterize each cluster.

Finally, Figure 5.3 shows the *intertopic distance map*, a two-dimensional visualization based on reduced embeddings. Clusters that appear close to each other share overlapping semantic content, whereas clusters that are well separated correspond to more distinct themes. Some overlap is visible, which indicates related arguments across clusters, while others are clearly distinct.

5.3 Agreement Ratio Analysis

In this section we report the agreement ratios for each of the eight selected topics, across models (GPT, LLaMA) and prompting conditions (role-based, native). Each figure shows how often the model's stance matched the gold stance for each country role, including the neutral assistant baseline.

Topic	Count	Name	Representation (top keywords)
4	114	Sex selection / women / gender	sex, selection, women, gender, parents, child, family, men, legalize, rape
3	117	Nuclear weapons / abolition / war	nuclear, weapons, abolition, war, power, energy, destruction, peace, world, fight
17	98	Prostitution / legalization	prostitution, legalizing, women, prostitutes, legalize, sex, profession, legalising, safer, trafficking
15	100	Economic sanctions / countries	sanctions, economic, countries, end, china, effective, country, hurt, citizens, trade
18	98	Multi-party political systems	party, multi, system, parties, political, systems, two, more, would, rti
22	89	Atheism / religion / belief	atheism, adopt, religion, believe, wars, religious, adopted, god, belief, beliefs
37	47	Compulsory voting / turnout	voting, compulsory, voter, turnout, vote, would, representative, election, voice, elections
40	42	Libertarianism / adopt / government	libertarianism, adopt, would, government, adopted, we, chaos, because, should, anarchy

Table 5.2: Selected BERTopic clusters: counts, readable names, and top c-TF–IDF keywords.

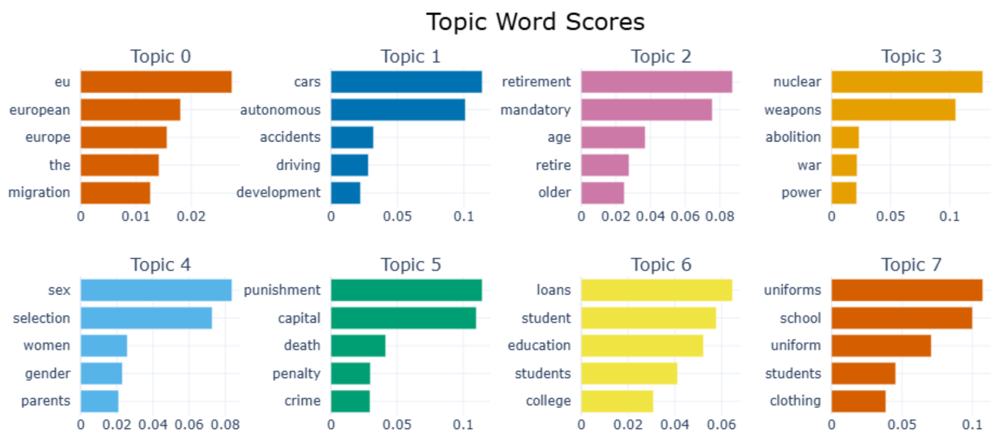

Figure 5.2: Topic word scores for selected clusters (higher bars indicate higher c-TF–IDF weight).

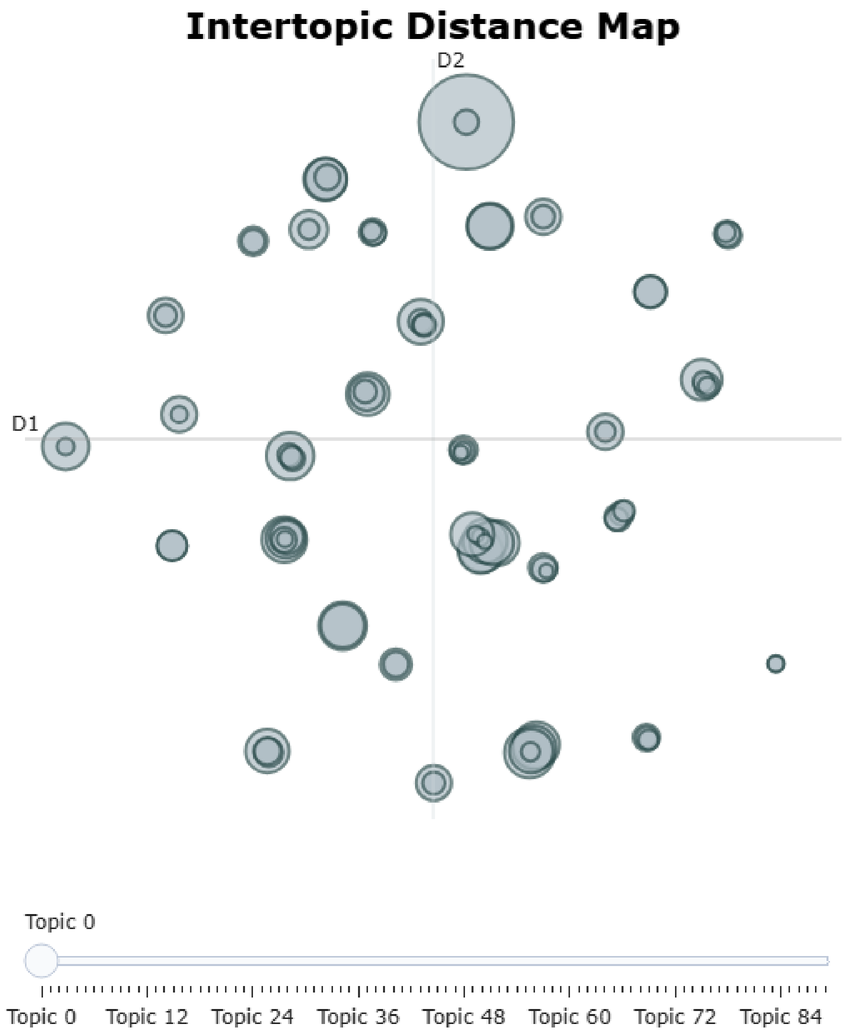

Figure 5.3: Intertopic distance map from BERTopic showing relative topic positions and sizes.

General patterns.

- The **assistant baseline** is generally stable across topics and often lies in the mid-range of agreement ratios.
- **GPT** tends to be more consistent across roles and languages, with smaller gaps between conditions.
- **LLaMA** shows larger variability: in some topics, its agreement ratios diverge widely across roles or between role-based and native prompts.
- The magnitude of the **role vs. native difference** is topic-dependent: for some (e.g., prostitution, multi-party system) the results are close, while for others (e.g., sanctions, atheism) the divergence is large.

Topic-wise observations.

- **No nuclear weapons**

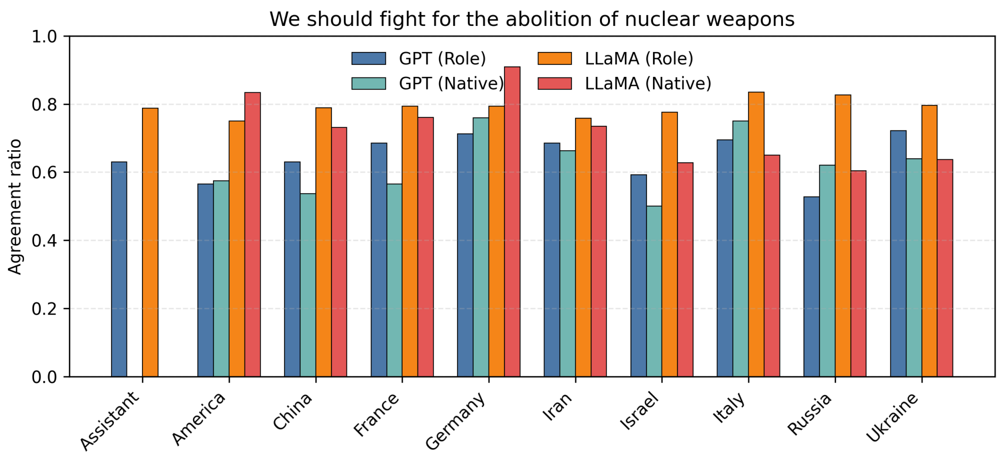

Figure 5.4: Agreement ratios by country for the topic "No nuclear weapons."

(Figure 5.4): GPT produces agreement ratios around 0.6–0.8 with limited role/native spread. LLaMA is more variable, especially for Iran and Germany. Role vs. native differences are moderate.

- **Legalize sex selection**

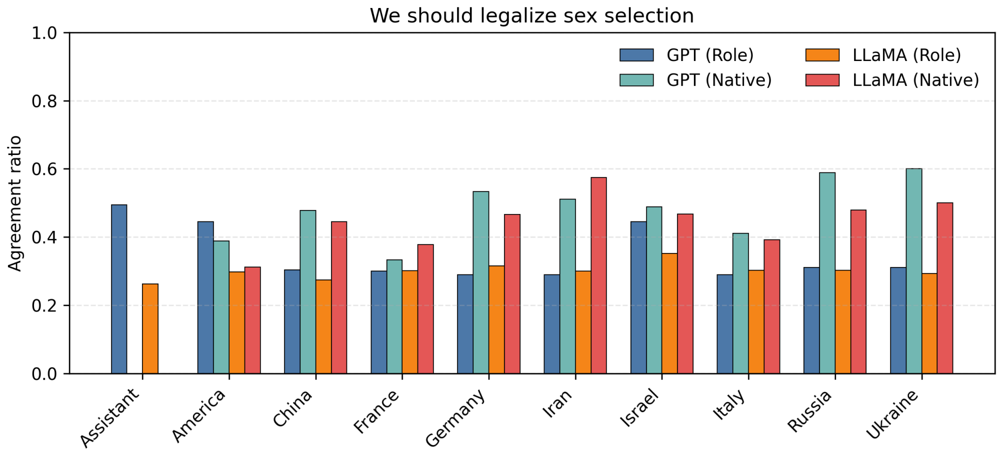

Figure 5.5: Agreement ratios by country for the topic “Legalize sex selection.”

(Figure 5.5): Both models lean toward disagreement (ratios often below 0.5). Larger role/native gaps appear in some countries (e.g., Iran, China).

- **No economic sanctions**

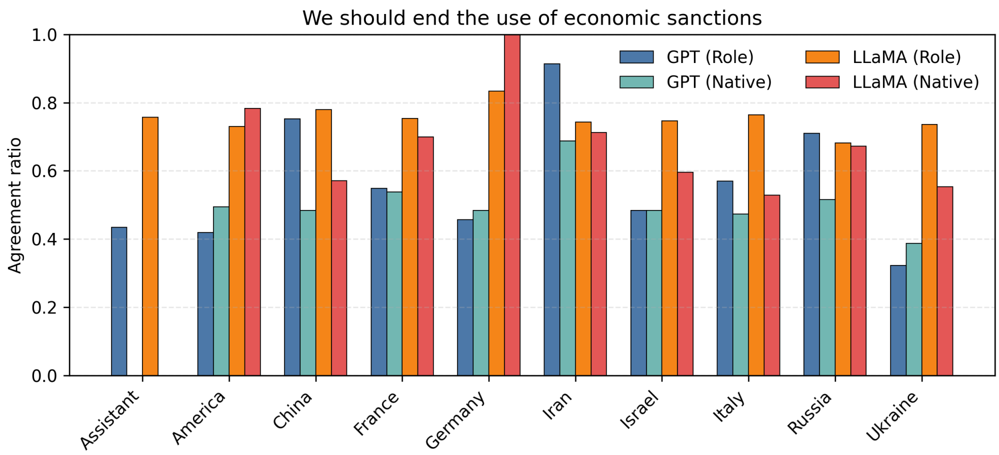

Figure 5.6: Agreement ratios by country for the topic “No economic sanctions.”

(Figure 5.6): Strong divergences. GPT remains in the 0.4–0.7 range, while LLaMA peaks close to 0.9 for Iran (role-based). Role/native gaps are particularly large in LLaMA.

- **Legalize prostitution**

(Figure 5.7): High agreement (> 0.7) across all models and conditions. Role vs. native differences are minimal.

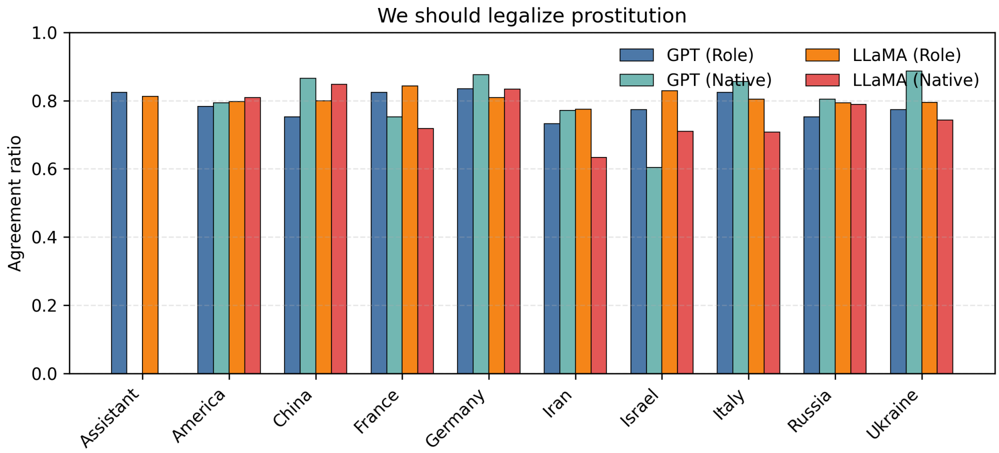

Figure 5.7: Agreement ratios by country for the topic “Legalize prostitution.”

- **Adopt multi-party system**

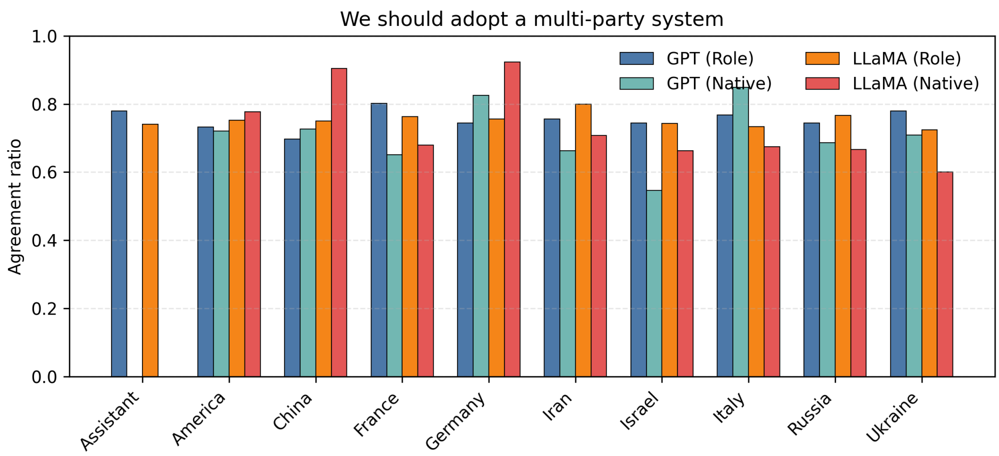

Figure 5.8: Agreement ratios by country for the topic “Adopt multi-party system.”

(Figure 5.8): Both models yield high agreement (0.7–0.9). Role and native completions are closely aligned, with little divergence.

- **Adopt atheism**

(Figure 5.9): Strong variation across roles and models. GPT tends to remain below 0.6, while LLaMA shows higher spread and larger role/native gaps. Divergences are particularly marked in countries with religious framing.

- **Compulsory voting**

(Figure 5.10): The lowest agreement ratios overall. GPT mostly falls around

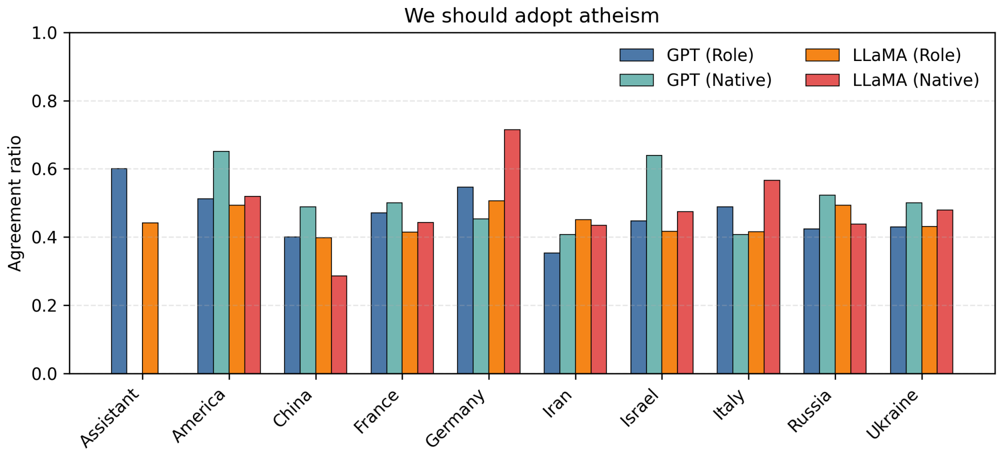

Figure 5.9: Agreement ratios by country for the topic "Adopt atheism."

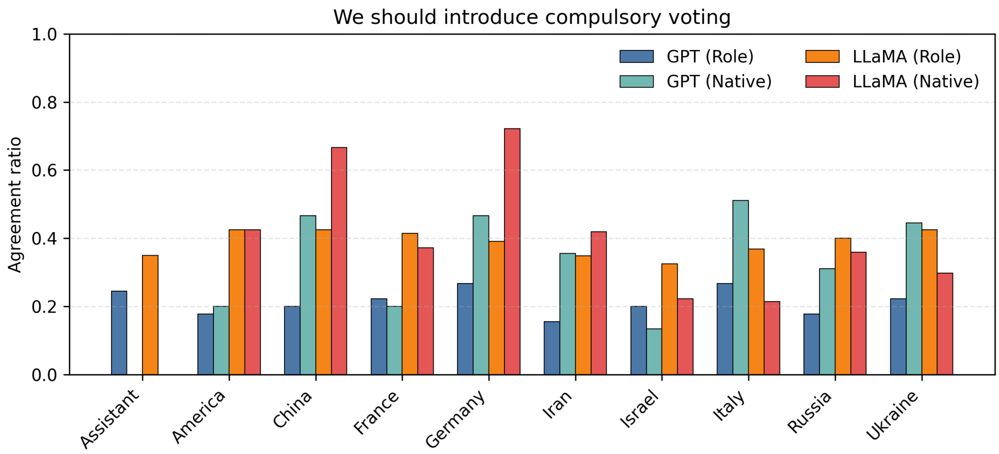

Figure 5.10: Agreement ratios by country for the topic "Compulsory voting."

0.2–0.4, while LLaMA ranges more widely, with some roles exceeding 0.6. Role/-native divergence is higher than in most other topics.

- **Adopt libertarianism**

  (Figure 5.11): Mid-range ratios (0.4–0.7). Both models show noticeable differences between role and native completions, though the size of the gap varies across countries.

Summary. Across topics, GPT demonstrates more stable agreement ratios with smaller role/native differences, while LLaMA exhibits stronger variability. Topics such as *prostitution* and *multi-party system* are consistently aligned across conditions, whereas *sanc-*

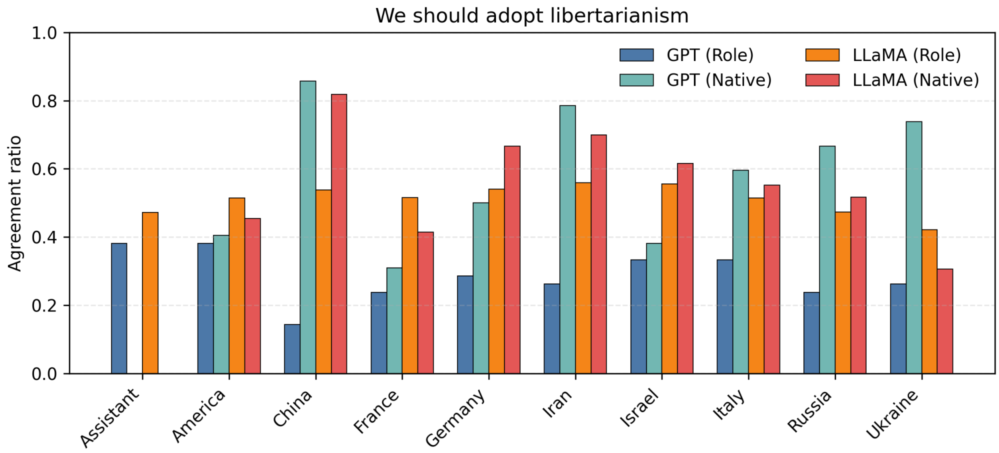

Figure 5.11: Agreement ratios by country for the topic "Adopt libertarianism."

tions, *atheism*, and *compulsory voting* show the largest divergences.

5.4 Embedding Similarity Results

In this section, we analyze the embedding similarities between conditions:

1. Role-based vs. Assistant responses,

2. Native vs. Role-based responses,

3. Native vs. Assistant responses.

Each bar represents the mean cosine similarity across topics for a given country, with error bars denoting the standard deviation.

A **higher similarity value** indicates that the two conditions being compared (e.g., role-based vs. assistant) produced embeddings that were closer in semantic space, suggesting that the model generated more similar reasoning structures across those prompting conditions. Conversely, a **lower similarity** means that the responses diverged more strongly in their semantic content, implying that the model adapted its reasoning depending on the prompt type.

A **small standard deviation** indicates that the model's behavior is stable across all topics, i.e., it shows a consistent degree of similarity between the compared conditions. A **large standard deviation**, on the other hand, implies greater variability: for some topics, the model produced very similar responses across conditions, while for others, the responses diverged more significantly. This variability is particularly informative as it points to topic-dependent sensitivity in the model's outputs.

5.4.1 GPT-4o-mini

Figure 5.12 shows that similarities between role-based and assistant responses remain relatively low across countries, generally centered around 0.55–0.6 with large standard deviations. Figure 5.13 highlights that native vs. role-based similarities are consistently higher, especially for America, China, and Italy, where scores approach 0.7. Figure 5.14 indicates that native vs. assistant similarities are the lowest of the three comparisons, with averages often near 0.5, suggesting stronger divergence between native-language completions and assistant completions.

5.4.2 LLaMA 3.1-70B

For LLaMA, results appear different. In Figure 5.15, role-based vs. assistant similarities are moderate (around 0.6–0.65), with relatively wide error bars for some countries (e.g., Iran, Ukraine). Figure 5.16 shows consistently higher similarities (0.65–0.8), particularly for America and France, which exhibit closer alignment between native and role-based answers. Finally, Figure 5.17 reveals that native vs. assistant similarities are comparable to the role-based vs. assistant case, again hovering near 0.65 for most countries.

5.4.3 General Patterns

Two clear trends emerge:

- Native vs. Role-based similarities are generally the highest across both models, suggesting these two prompting strategies produce the most consistent outputs.
- Native vs. Assistant similarities tend to be lowest, implying that the "helpful assistant" completions diverge more strongly from role-based or native completions.

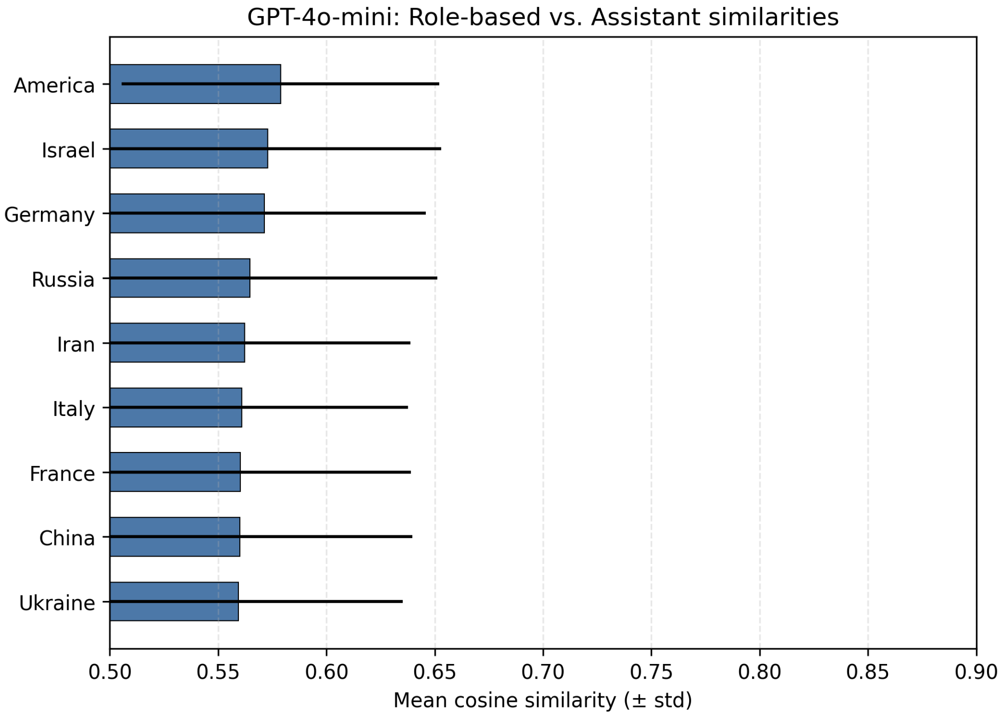

Figure 5.12: GPT-4o-mini: Role-based vs. Assistant similarities.

5.5 Model Comparison (GPT vs LLaMA)

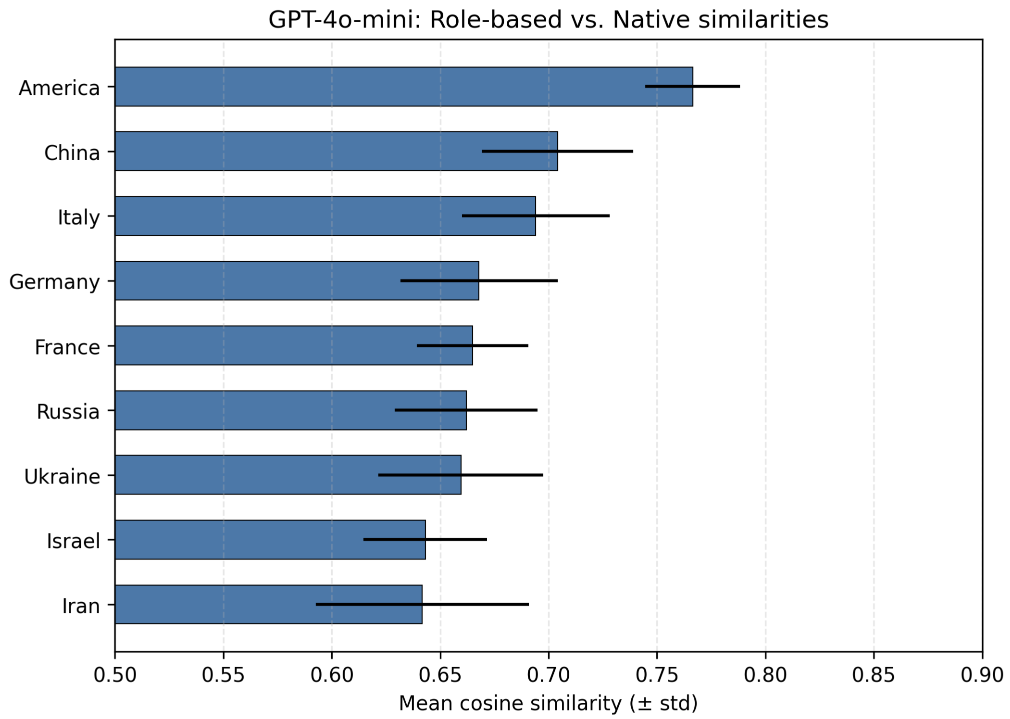

Figure 5.13: GPT-4o-mini: Native vs. Role-based similarities.

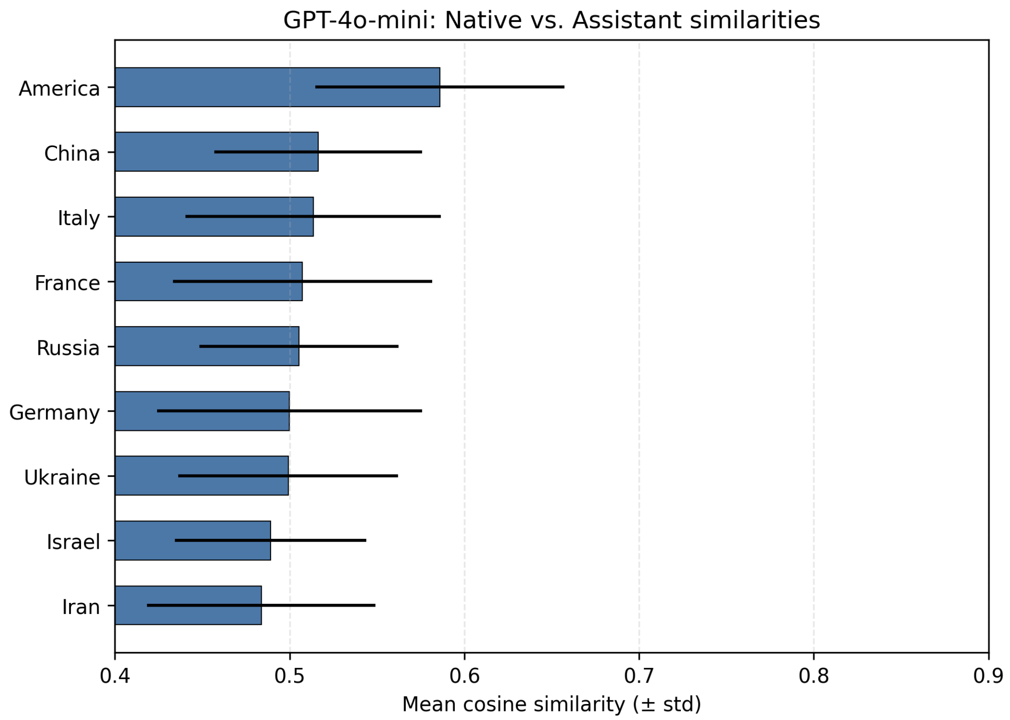

Figure 5.14: GPT-4o-mini: Native vs. Assistant similarities.

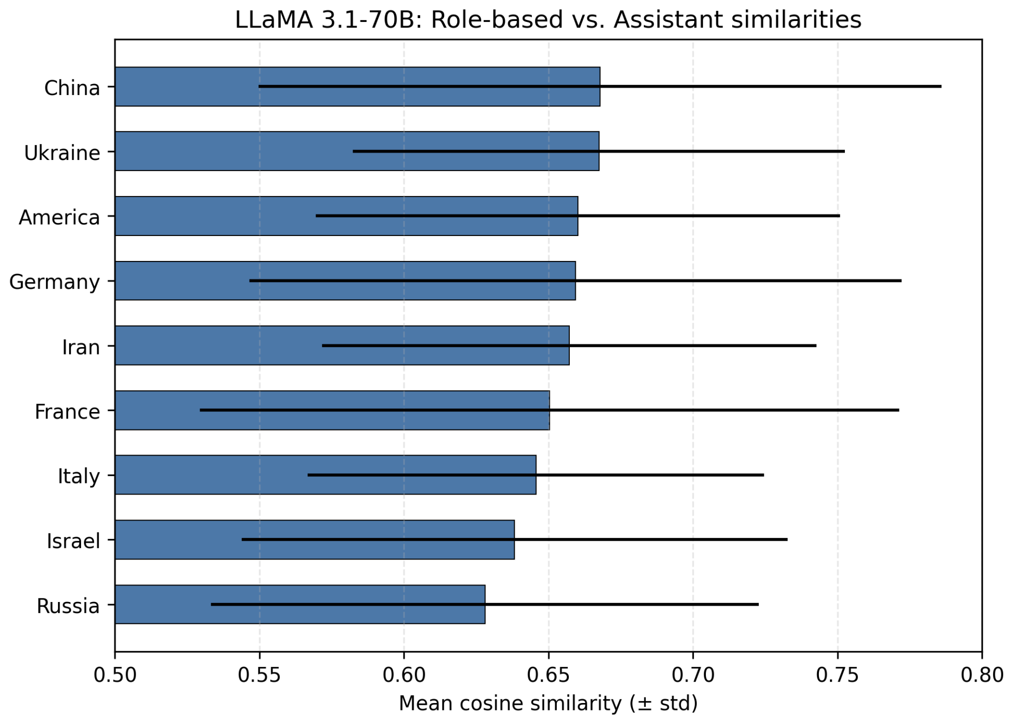

Figure 5.15: LLaMA 3.1-70B: Role-based vs. Assistant similarities.

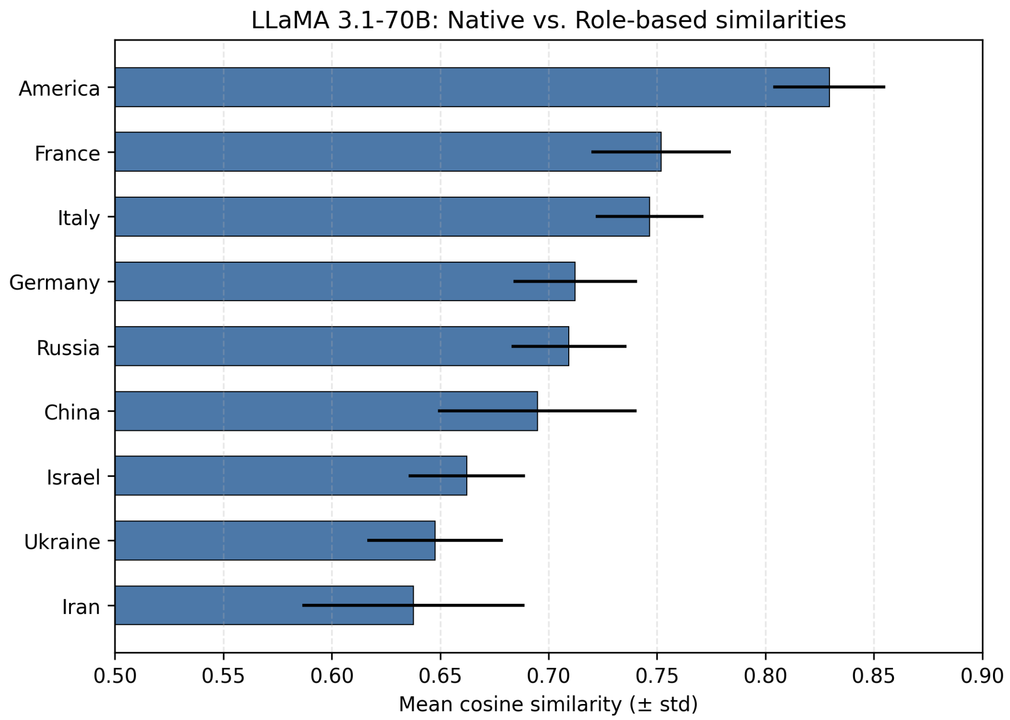

Figure 5.16: LLaMA 3.1-70B: Native vs. Role-based similarities.

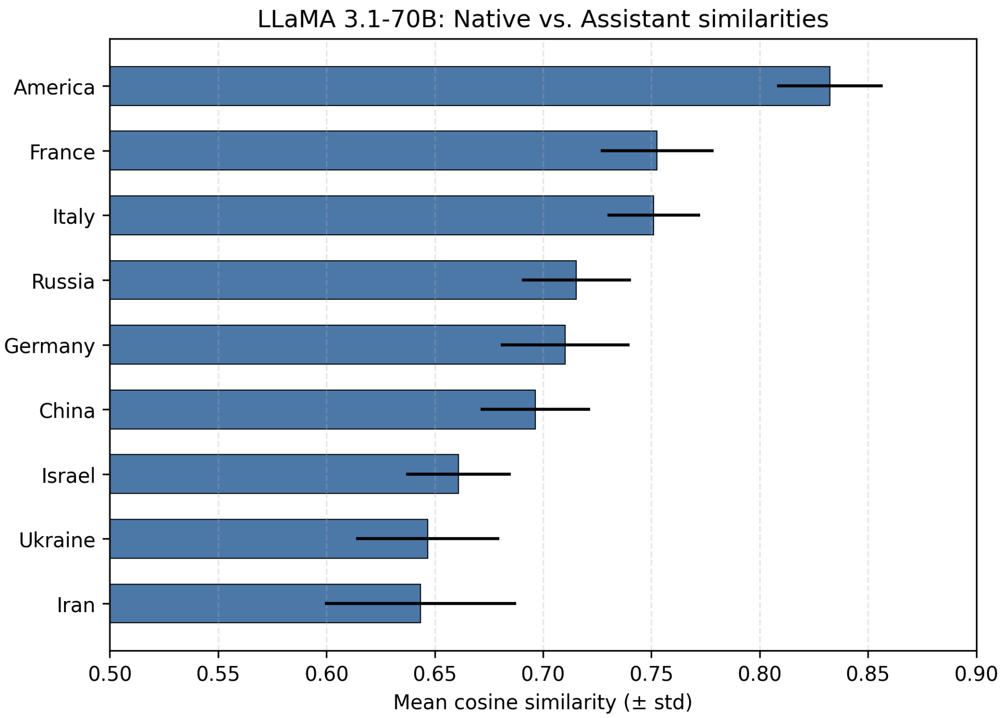

Figure 5.17: LLaMA 3.1-70B: Native vs. Assistant similarities.

Chapter 6

Limitations and Future works

References

[1] OpenAI et al. *GPT-4 Technical Report*. 2023. arXiv: 2303.08774 [cs.CL]. URL: https://arxiv.org/abs/2303.08774.

[2] Hugo Touvron et al. *LLaMA: Open and Efficient Foundation Language Models*. 2023. arXiv: 2302.13971 [cs.CL]. URL: https://arxiv.org/abs/2302.13971.

[3] Laura Weidinger et al. *Ethical and Social Risks of Harm from Language Models*. 2021. arXiv: 2112.04359 [cs.CL]. URL: https://arxiv.org/abs/2112.04359.

[4] Shibani Santurkar et al. "Whose Opinions Do Language Models Reflect?" In: *Proceedings of the 40th International Conference on Machine Learning*. Ed. by Andreas Krause et al. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 29971–30004. URL: https://proceedings.mlr.press/v202/santurkar23a.html.

[5] Junfeng Jiao et al. *Navigating LLM Ethics: Advancements, Challenges, and Future Directions*. 2025. arXiv: 2406.18841 [cs.CY]. URL: https://arxiv.org/abs/2406.18841.

[6] Keita Kurita et al. *Measuring Bias in Contextualized Word Representations*. 2019. arXiv: 1906.07337 [cs.CL]. URL: https://arxiv.org/abs/1906.07337.

[7] Moin Nadeem, Anna Bethke, and Siva Reddy. *StereoSet: Measuring stereotypical bias in pretrained language models*. 2020. arXiv: 2004.09456 [cs.CL]. URL: https://arxiv.org/abs/2004.09456.

[8] Md Abdul Aowal et al. *Detecting Natural Language Biases with Prompt-based Learning*. 2023. arXiv: 2309.05227 [cs.CL]. URL: https://arxiv.org/abs/2309.05227.

[9] Nicole Gross. “What ChatGPT Tells Us about Gender: A Cautionary Tale about Performativity and Gender Biases in AI”. In: *Social Sciences* 12.8 (2023). ISSN: 2076-0760. DOI: 10.3390/socsci12080435. URL: https://www.mdpi.com/2076-0760/12/8/435.

[10] Aleksandra Urman and Mykola Makhortykh. “The silence of the LLMs: Cross-lingual analysis of guardrail-related political bias and false information prevalence in ChatGPT, Google Bard (Gemini), and Bing Chat”. In: *Telematics and Informatics* 96 (2025), p. 102211. ISSN: 0736-5853. DOI: https://doi.org/10.1016/j.tele.2024.102211. URL: https://www.sciencedirect.com/science/article/pii/S0736585324001151.

[11] Pranav Narayanan Venkit et al. *Nationality Bias in Text Generation*. 2023. arXiv: 2302.02463 [cs.CL]. URL: https://arxiv.org/abs/2302.02463.

[12] Richard Rogers and Xiaoke Zhang. “The Russia–Ukraine War in Chinese Social Media: LLM Analysis Yields a Bias Toward Neutrality”. In: *Social Media + Society* 10 (May 2024). DOI: 10.1177/20563051241254379.

[13] Nailia Mirzakhmedova et al. *The Touché23-ValueEval Dataset for Identifying Human Values behind Arguments*. 2023. arXiv: 2301.13771 [cs.CL]. URL: https://arxiv.org/abs/2301.13771.

[14] Nikita Soni et al. “Human Language Modeling”. In: *Findings of the Association for Computational Linguistics: ACL 2022*. Ed. by Smaranda Muresan, Preslav Nakov, and Aline Villavicencio. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 622–636. DOI: 10.18653/v1/2022.findings-acl.52. URL: https://aclanthology.org/2022.findings-acl.52/.

[15] Maarten Grootendorst. *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*. 2022. arXiv: 2203.05794 [cs.CL]. URL: https://arxiv.org/abs/2203.05794.

[16] Nikita Nangia et al. *CrowS-Pairs: A Challenge Dataset for Measuring Social Biases in Masked Language Models*. 2020. arXiv: 2010.00133 [cs.CL]. URL: https://arxiv.org/abs/2010.00133.

[17] Fangxiaoyu Feng et al. *Language-agnostic BERT Sentence Embedding*. 2022. arXiv: 2007.01852 [cs.CL]. URL: https://arxiv.org/abs/2007.01852.

[18] Tomas Mikolov et al. *Efficient Estimation of Word Representations in Vector Space*. 2013. arXiv: 1301.3781 [cs.CL]. URL: https://arxiv.org/abs/1301.3781.

[19] Jeffrey Pennington, Richard Socher, and Christopher Manning. "GloVe: Global Vectors for Word Representation". In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Ed. by Alessandro Moschitti, Bo Pang, and Walter Daelemans. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1532–1543. DOI: 10.3115/v1/D14-1162. URL: https://aclanthology.org/D14-1162/.

[20] Matthew E. Peters et al. *Deep contextualized word representations*. 2018. arXiv: 1802.05365 [cs.CL]. URL: https://arxiv.org/abs/1802.05365.

[21] Jacob Devlin et al. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. 2019. arXiv: 1810.04805 [cs.CL]. URL: https://arxiv.org/abs/1810.04805.

[22] Alec Radford et al. *Improving language understanding by generative pre-training*. 2018.

[23] Tom B. Brown et al. *Language Models are Few-Shot Learners*. 2020. arXiv: 2005.14165 [cs.CL]. URL: https://arxiv.org/abs/2005.14165.

[24] Hugo Touvron et al. *Llama 2: Open Foundation and Fine-Tuned Chat Models*. 2023. arXiv: 2307.09288 [cs.CL]. URL: https://arxiv.org/abs/2307.09288.

[25] Ashish Vaswani et al. *Attention Is All You Need*. 2023. arXiv: 1706.03762 [cs.CL]. URL: https://arxiv.org/abs/1706.03762.

[26] Long Ouyang et al. *Training language models to follow instructions with human feedback*. 2022. arXiv: 2203.02155 [cs.CL]. URL: https://arxiv.org/abs/2203.02155.

[27] Bernd Carsten Stahl and Damian Eke. *The ethics of ChatGPT – Exploring the ethical issues of an emerging technology*. 2024. DOI: https://doi.org/10.1016/j.ijinfomgt.2023.102700. URL: https://www.sciencedirect.com/science/article/pii/S0268401223000816.

[28] Patrick Haller, Ansar Aynetdinov, and Alan Akbik. *OpinionGPT: Modelling Explicit Biases in Instruction-Tuned LLMs*. 2023. arXiv: 2309.03876 [cs.CL]. URL: https://arxiv.org/abs/2309.03876.

[29] Anthony G. Greenwald, Debbie E. McGhee, and Jordan L. K. Schwartz. "Measuring Individual Differences in Implicit Cognition: The Implicit Association Test". In: *Journal of Personality and Social Psychology* 74.6 (1998), pp. 1464–1480. DOI: 10.1037/0022-3514.74.6.1464. URL: https://psycnet.apa.org/doi/10.1037/0022-3514.74.6.1464.

[30] Chandler May et al. *On Measuring Social Biases in Sentence Encoders*. 2019. arXiv: 1903.10561 [cs.CL]. URL: https://arxiv.org/abs/1903.10561.

[31] Paula Czarnowska, Yogarshi Vyas, and Kashif Shah. "Quantifying Social Biases in NLP: A Generalization and Empirical Comparison of Extrinsic Fairness Metrics". In: *Transactions of the Association for Computational Linguistics* 9 (2021). Ed. by Brian Roark and Ani Nenkova, pp. 1249–1267. DOI: 10.1162/tacl_a_00425. URL: https://aclanthology.org/2021.tacl-1.74/.

[32] Nils Reimers and Iryna Gurevych. *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*. 2019. arXiv: 1908.10084 [cs.CL]. URL: https://arxiv.org/abs/1908.10084.

[33] Abubakar Abid, Maheen Farooqi, and James Zou. “Persistent Anti-Muslim Bias in Large Language Models”. In: *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*. ACM, 2021, pp. 298–306. DOI: 10.1145/3461702.3462624. URL: https://doi.org/10.1145/3461702.3462624.

[34] Eddie Yang and Margaret E. Roberts. “Censorship of Online Encyclopedias: Implications for NLP Models”. In: *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’21)*. ACM, 2021, pp. 298–306. DOI: 10.1145/3442188.3445916. URL: https://doi.org/10.1145/3442188.3445916.

[35] Yupei Du, Johnny Tian-Zheng Wei, and Heng Ji. “The Impact of Online Censorship on Language Models”. In: *Proceedings of the First Workshop on Gender Bias in Natural Language Processing*. Association for Computational Linguistics, 2019, pp. 80–88. DOI: 10.18653/v1/D19-1635. URL: https://aclanthology.org/D19-1635.

[36] Jieyu Zhao et al. “Gender Bias in Coreference Resolution: Evaluation and Debiasing Methods”. In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, 2018, pp. 15–20. DOI: 10.18653/v1/N18-2002. URL: https://aclanthology.org/N18-2002.

[37] Kellie Webster et al. “Mind the GAP: A Balanced Corpus of Gendered Ambiguous Pronouns”. In: *Transactions of the Association for Computational Linguistics*. Vol. 6. 2018, pp. 605–617. DOI: 10.1162/tacl_a_00041. URL: https://aclanthology.org/Q18-1041.

[38] Jwala Dhamala et al. “BOLD: Dataset and Metrics for Measuring Biases in Open-Ended Language Generation”. In: *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (2021), pp. 862–872. DOI: 10.1145/3442188.3445924. URL: https://arxiv.org/abs/2101.11718.

[39] OpenAI. *Learn how to use JSON mode*. https://platform.openai.com/docs/guides/json-mode. Accessed: 2025-07-02. 2025.

[40] Nidhal Baccouri. *deep-translator: A flexible free and unlimited python tool to translate between different languages in a simple way using multiple translators.* https://pypi.org/project/deep-translator/. Version 1.11.4, accessed: 2025-08-24. 2020.